Alan Wrya

Toronto, ON | alan.wrya@torontomu.ca | (416) 939-5417 | linkedin.com/in/alan-w | github.com/left-nullspace

Summary

Final-year Financial Mathematics Co-op student passionate about trading, financial markets, and quantitative research. Hands-on experience across bank-side analytics (TD Direct Investing), portfolio research (cluster-constrained MV optimization), and trading-software development (Anomalies platform).

Education

Toronto Metropolitan University
Sept 2022 – June 2026

Bachelor of Science (BSc), Financial Mathematics Co-op | Computer Science Minor
Toronto, ON

- **CGPA: 3.81** | Dean's List Scholar
- *Relevant Coursework:* Stochastic Calculus, Partial Differential Equations, Investment Analysis, Managerial Finance, Probability & Statistics, Linear Algebra, Differential Equations, Data Structures, Computer Science

Experience

TD Bank
Jan 2025 – Aug 2025

Co-op, Direct Investing Journey Analyst
Toronto, ON

- Rebuilt 12+ monthly KPI dashboards in Excel using VBA, LAMBDA functions, dynamic named ranges, and reusable chart templates, reducing manual refresh time from 2-5 hours per week to under 5 minutes per dashboard.
- Engineered automated 24-month rolling performance charts with dynamic labels and auto-scaling axes, enabling leadership to refresh reports on demand and standardize reporting across the team.
- Built client segmentation frameworks and ran correlation and regression analysis on three years of monthly client data to surface drivers of retention, funding, and engagement.
- Led analytics tagging plans for new Direct Investing features: reviewed wireframes, scoped high-value events, presented to tagging and analytics teams, and coordinated risk sign-off across product, analytics, and 1B controls.
- Ran feature-launch lookback analyses on enhanced search adoption and mobile vs. desktop trading behaviour in Adobe Analytics and Excel to inform roadmap prioritization and executive-facing recommendations.
- Took on expanded analytics, stakeholder, and reporting responsibilities during the manager's parental leave.

Ted Rogers Sales & Trading (TRST)
Sept 2023 – Dec 2025

Quantitative Analyst
Toronto, ON

- Built a Portfolio Optimization & Simulation tool using historical data and mean-variance optimization to generate optimal Sharpe and minimum-volatility portfolios.
- Developed a Value-at-Risk tool using Monte Carlo simulations to assess potential portfolio losses from a distribution of outcomes.
- Adapted an interactive Black-Scholes option pricing calculator to compute call/put prices, implied volatility, and probabilities of expiring in-the-money.

R&R Utility Limited
Sept 2021 – Apr 2024

Project Management Assistant
Toronto, ON

- Supported the development of corporate operating budgets, project costing, and drawdown budgets across multi-discipline engineering and construction projects.
- Ran financial reconciliations on vendor invoices to resolve billing discrepancies and improve cost-tracking accuracy.

Research & Projects

Cluster-Constrained Portfolio Optimization
2025 – 2026

Python, Scikit-learn

- Researched whether sector-clustering constraints improve mean-variance portfolio performance vs. plain MV on a 33-stock sector-stratified universe.
- Built an end-to-end backtesting pipeline: Ward hierarchical clustering → cluster-constrained MV optimization → benchmarking against Plain MV and ERC across 860 rolling 5-year windows.

- Showed cluster-constrained MV matches plain MV's out-of-sample Sharpe while materially lowering portfolio concentration, max drawdown, and 5% expected shortfall.
- Identified that plain MV behaves as an *error maximizer*, concentrating allocations in in-sample winners; cluster constraints mitigate this without sacrificing risk-adjusted returns.

Anomalies: Point-and-Click Trading Strategy Discovery Platform — 2025 – Present

Python, DEAP, vectorbtpro, PySide6

- Core builder on a small team developing a point-and-click desktop application that lets users design, backtest, and stress-test systematic trading strategies across crypto and equity markets.
- Designed and built the backtesting engine, stress-testing pipeline, and evolutionary search system (DEAP) that discovers novel strategies by exploring combinations of technical-indicator signals and exit parameters.
- Built the desktop UI (PySide6) and a multi-source market data layer (Yahoo Finance, Binance, dYdX, Paradex, Hyperliquid) with local SQLite caching and Turso cloud sync.

Skills

Programming | Python (Pandas, NumPy, Scikit-learn, Statsmodels), SQL, MATLAB, R, Java, VBA
Quantitative & Finance | Mean-variance optimization, Monte Carlo simulation, Black-Scholes, Value-at-Risk, backtesting, time-series analysis, clustering, PCA
Tools | Adobe Analytics, Excel (LAMBDA, dynamic ranges, VBA macros), PowerPoint, Git, Jupyter
AI / Agentic Workflows | Claude, Codex, GitHub Copilot for agentic code generation, research, and automation

Activities

Math Tutor (Self-Employed) — Sept 2022 – Present

Private High-School Mathematics Instruction — Toronto, ON

- One-on-one tutoring in high-school calculus, algebra, trigonometry, and geometry; built customized lesson plans tailored to each student's learning style and academic goals.